

Galaxy evolution as a function of cosmic time

Evolution of star formation rate vs stellar mass in redshift

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ABSTRACT

Aims. The aim is to study the relation between star formation rate versus stellar mass for a sample of about 20,000 galaxies from CANDELS+3D-HST catalog. In particular the galaxies are divided into 4 different redshift z bins and for each of them the relation is studied. In addition, the fraction of passive and active galaxies is computed in the considered redshifts.

Methods. Using the data of spectroscopy and photometry, the galaxies are divided into 4 redshift bins. From the relation in the plane UV-VJ the active and passive galaxies are separated. Then from a fit of the SFR vs stellar mass for the active galaxies, the SFR evolution in redshift is obtained.

Results. The expected relations and evolutions are obtained.

Key words. SFR , $SSFR$, z , passive and active fractions

1. Introduction

In the past decade, the presence of more and more accurate multi-wavelength telescopes has led to new insights of galaxies and their evolution. One of the studies shows the behavior of stellar mass connected with star formation rate SFR . In fact, by putting a relation between these quantities, it can be described how efficiently a galaxy can transform gas into stars and it can be analyzed the presence of particular physical processes (ex. feedback mechanisms).

Galaxies can be divided into passive or active from the value of their SFR . One of the important contributions in the analysis of these fractions was given by Salim et al. (2007). Given their plot (Figure 1) of specific star formation rate $SSFR$ vs stellar mass, it can be concluded that on average smaller galaxies are active, while bigger galaxies are passive and are hosting an AGN (Active galactic nucleus with radio feedback and AGN feedback are responsible for lowering the SFR).

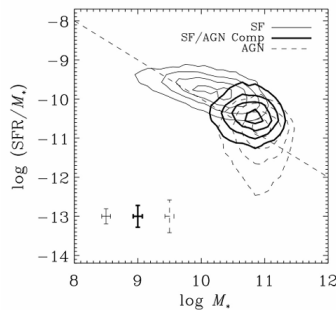


Fig. 1. Figure 19 in Salim et al. (2007): Specific SFR versus stellar mass, making explicit the galaxies hosting an AGN.

Afterwards, thanks to Whitaker et al. (2011)'s work, the SFR was correlated to the color. In their paper, instead of using directly the SFR , the fraction of passive and active galaxies is found as a bimodal distribution in the plane U-V and V-J (Fig-

ure 2). From here, they also obtain a criteria to divide the two populations.

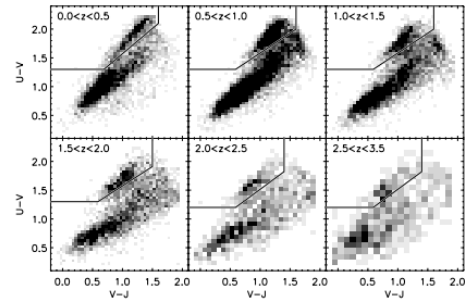


Fig. 2. Figure 17 in Whitaker et al. (2011): Rest-frame U-V vs V-J diagram for NMBS galaxies.

The relation between $SSFR$ and stellar mass has had another important contribution given by Muzzin et al. (2013). In particular they analyzed the passive fraction in relation to the stellar mass and the evolution with time (Figure 3). They found that the higher the z is, the fewer and on average more massive the passive galaxies are; while, with the decrease of the redshift, passive galaxies are higher in number and also present at smaller masses.

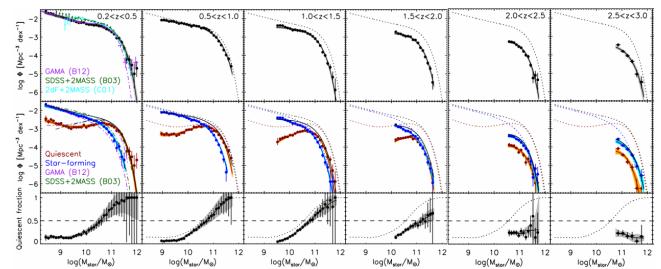


Fig. 3. Figure 6 in Muzzin et al. (2013): Fraction of quiescent or star-forming galaxies and stellar mass in different redshift bins.

Lastly, an important advancement in galaxy evolution was made by [Madau and Dickinson \(2014\)](#) with the SFR vs redshift study. From their plot (Figure 4) it can be seen that at higher and lower redshift the SFR is smaller, while there is a maximum at the cosmic noon ($z = 2$). So, from this analysis it is expected that the redshift bin with $z = 2$ has the highest SFR , while the other ones are smaller. This difference in the SFR comes from two principal processes: for the high redshift, the presence of small perturbation and so smaller halos cause smaller star formation. While for lower redshift, the presence of feedback processes decreases the SFR in galaxies.

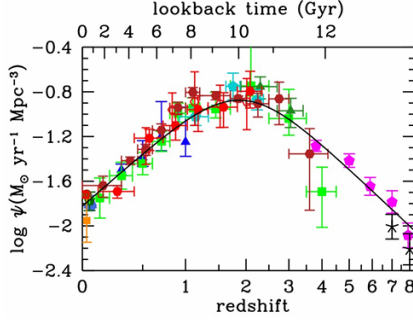


Fig. 4. Figure 9 in [Madau and Dickinson \(2014\)](#): SFR over volume and redshift.

2. Datasets

The galaxies utilized in this work are from the Cosmic Assembly Near-IR Deep Extragalactic Legacy Survey (CANDELS)+3D-Hubble Space Telescope (3D-HST). These programs, thanks to observations of several ground- and space-based telescopes, obtained photometry and spectroscopy of about 20,000 sources distributed in five areas of the sky.

In particular CANDELS (912-orbit Multi-Cycle Treasury imaging program) is the largest imaging survey done with WFC3, a camera on the Hubble Space Telescope. It focuses on the observation of objects in the first third of galactic evolution from $z = 8$ to 1.5 in five extragalactic fields: All-wavelength Extended Groth Strip International Survey (AEGIS) field, Cosmic Evolution Survey (COSMOS) field, Great Observatories Origins Survey (GOODS) Northern and Southern fields (GOODS-North and GOODS-South) and UKIRT InfraRed Deep Sky Surveys (UKIDSS) Ultra Deep Field (UDS) (Figure 5). The CANDELS results cover wavelengths from near-ultraviolet to the far-infrared and are also combined with Hubble Ultradeep Field program so that the sample is more complete.

3D-HST (248-orbit HST Treasury program), instead, is a near-infrared spectroscopic survey. This program studies galaxies (with physical processes and shape) of the distant Universe using WFC3 G141. In particular, it provides a surveying of about 600 square arcminutes, 75% of the CANDLE area, with redshift z between 1 and 3 as third dimension.

So, CANDELS+3D-HST combined give a total surveying of over ~900 square arcminutes and are optimal to study galaxies at redshift between 0.5 and 3.5, as it is done in this work. An important note on the dataset utilized is that some of the data presents values out of the evaluated physical range. So these galaxies are not considered and analyzed.

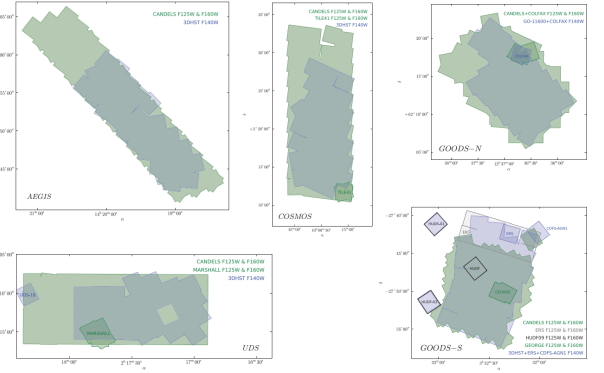


Fig. 5. [Skelton et al. \(2014\)](#): Layout of the WFC3 observations, CANDELS+3D-HST.

3. Results

The first thing done in this work is to analyze the passive and active galaxies in the U-V and V-J color plane, following the [Whitaker et al. \(2011\)](#). The distribution obtained is a bimodal one (Figure 3): in the bigger cloud is expected to be the main sequence of galaxy evolution, so with active galaxies; while the smaller cloud should contain the passive ones. This is checked by using the criteria derived by [Whitaker et al. \(2011\)](#) (Figure 3):

$$\begin{cases} (U - V) > 0.88 \times (V - J) + 0.59, \\ (U - V) > 1.3, \\ (V - J) < 1.6. \end{cases} \quad (1)$$

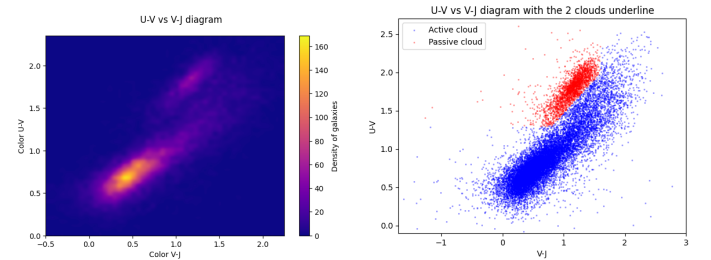


Fig. 6. Left-hand panel: U-V versus V-J plane for the considered galaxies. Right-hand panel: Division in passive and active clouds in U-V versus V-J plane.

Then, the galaxies are divided into 4 redshift bins: $z_1=[0.6, 1]$, $z_2=[1, 1.7]$, $z_3=[1.7, 2.3]$ and $z_4=[2.3, 2.9]$. For these bins the U-V and V-J plane is filled. As can be seen in Figure 7, an evolution of the SFR of these galaxies is occurring. In the higher redshift, so when the Universe was young, most of the galaxies are at the begging of their evolution and are at the top of the main sequence. As the redshift goes down, a cloud of passive galaxies grows.

The division passive/active is then checked in two ways: by coloring the U-V vs V-J diagram with the values of $SSFR$ (expressed in $1/yr$) and by using the same plot as [Salim et al. \(2007\)](#) (Figure 1). In the first verification the $SSFR$ is used and not the SFR to avoid making a mistake in the analysis of smaller galaxies: their SFR can appear small and then be considered passive even if they are active, while with the $SSFR$ this does not occur. In the plot (Figure 3), the two clouds are stained with different

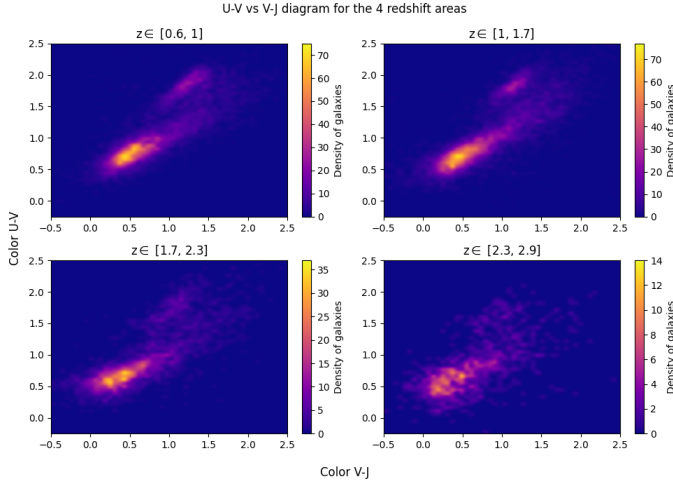


Fig. 7. U-V versus V-J plane for the four redshift bins considered.

colors indicating two different values of $SSFR$: the expected active cloud has a bigger value of $SSFR$, while the passive one presents smaller values. In the second verification, the galaxies are plotted in a $SSFR$ (expressed in $1/\text{yr}$) vs stellar mass (expressed in $M_\odot$) plane (Figure 3). In this case as well, the two populations can be seen: in particular the passive ones are on average more massive than the active ones. This phenomenon is called downsizing: small halos collapse first but they need time to accrete mass in order to start star formation. Instead, bigger halos collapse later on but they immediately begin with star formation, then the feedback processes activate bringing these massive galaxies in a passive mode. This explains why passive galaxies are more massive than the active ones.

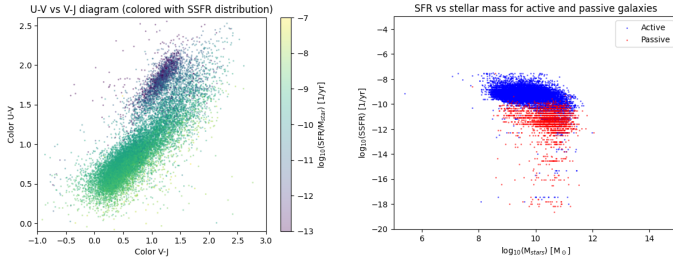


Fig. 8. Left-hand panel: U-V vs V-J plane for all the galaxies colored with their $SSFR$. Right-hand panel: $SSFR$ vs stellar mass plane for all galaxies divided in the two clouds.

Then, the evolution in redshift of the SFR -mass diagram is analyzed. Given the redshift bins considered before, for each of them the SFR -mass plot is computed (Figure 11). In all of them the main structure is the same, but there is a little difference in the passive cloud: at higher redshift it appears small, while at lower z it is bigger.

To better explore the redshift dependence in the SFR , for the z bins, the SFR vs mass relation is considered for only the active galaxies. In particular, the aim is to obtain a plot of how the SFR evolves with redshift respect to stellar mass. The choice of only using the active galaxies is because nearly all the star formation occurs in these galaxies and they almost present a linear relation between SFR and stellar mass. So, given the distribution of active galaxies, a linear fit has to be computed. However the data presents a scatter towards the passive galaxies. This dispersion

is due to the criteria used to obtain the fractions: the division using UV-VJ quantities is not accurate; for the passive galaxies is a good relation, but as regard as the active some of them are too near the passive ones. To avoid this problem the data can be cut out, considering only the main cloud and not the scattered points. By doing this and also using the $SSFR$ instead of the SFR , the fit of the cloud improves (Figure 12).

The obtained data covers a redshift interval between 0.6 and 2.9. In order to complete the dataset with $z = 0$, the fit of active galaxies in Bluck et al. (2019) is used (Figure 9).

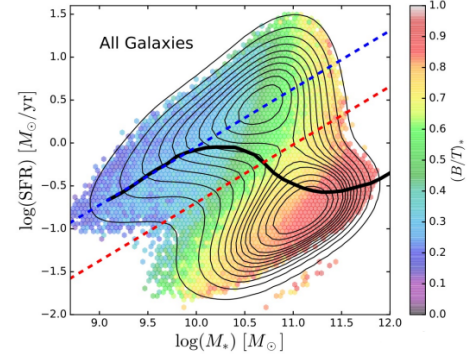


Fig. 9. Figure 1 in Bluck et al. (2019): star-forming main sequence and the passive cloud for SDSS galaxies ($z \sim 0.1$).

All obtained parameters (Table 1) for every considered redshift are then shown in a $SSFR$ -mass diagram (Figure 10). In this case as well, there is a clear evolution of the $SSFR$ values: at higher redshift the SFR was bigger than the one at smaller z . The third redshift interval however does not follow this evolution. This bin contains $z = 2$, the cosmic noon, that corresponds to the SFR peak, so the SFR is bigger and approximately constant for all masses.

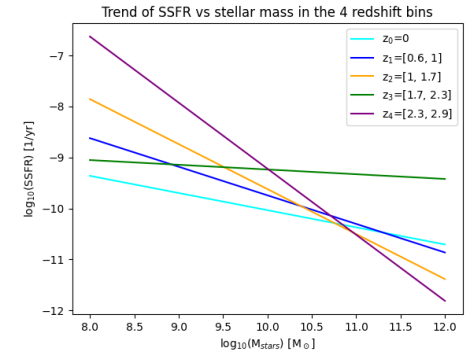


Fig. 10. Trends of the fits found in $SSFR$ vs stellar mass plane.

The different values of SFR came from various processes. At higher z the gas was denser and so there was more SF. Then, with the expansion of the Universe, the density went down, and, in particular, from the cosmic noon the accretion mode changed: from cold accretion with streams to a hot mode accretion, that is less efficient, leading the SFR to drop. Another important change that reduced the SFR was the beginning of feedback processes: from all the massive stars formed, the supernovae feedback occurs; thanks to all the mass accreted in the galaxies, the AGN feedbacks activates. So, the SFR at $z = 0$ is smaller as it can be seen from Figure 10. There is still some SF in the smaller clouds, while the massive galaxies are for the most passive.

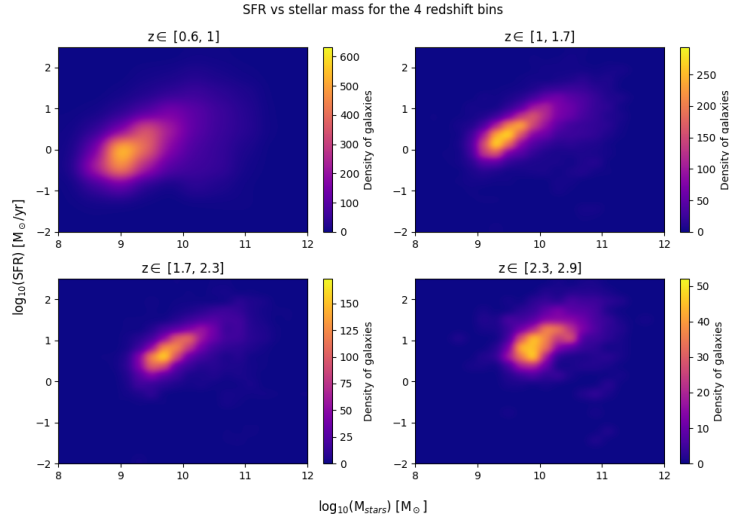


Fig. 11. *SFR* vs stellar mass plane for the active galaxies (all data) in the 4 redshift bins.

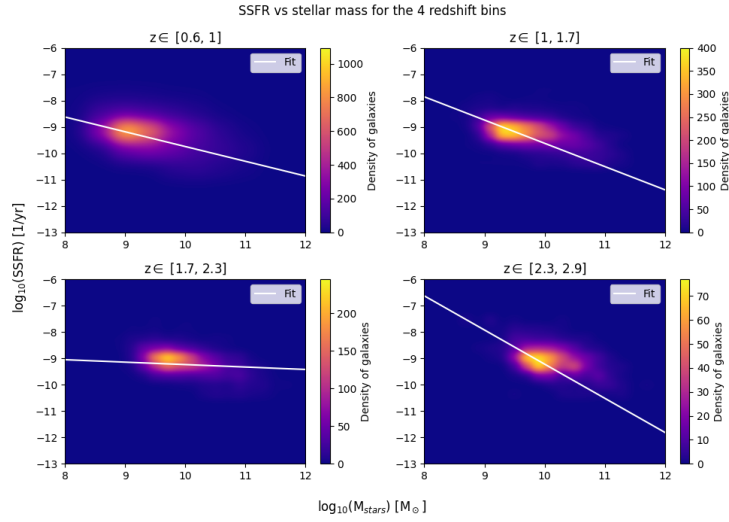


Fig. 12. *SSFR* vs stellar mass plane for the active galaxies (cut data) in the 4 redshift bins with correspondent fits.

	m	q
z_0	$-0.336454 \pm 3.6e-6$	$-6.6696 \pm 3.8e-4$
z_1	-0.561 ± 0.006	-4.138 ± 0.499
z_2	-0.882 ± 0.007	-0.799 ± 0.683
z_3	$-0.092564 \pm 1.9e-5$	-8.311 ± 0.002
z_4	-1.296 ± 0.102	3.73560 ± 10.48

Table 1. Values of the parameters m and q obtained for fits in the redshift bins.

4. Conclusions

All the results obtained are coherent with the literature. In particular in the first part, the same results of [Whitaker et al. \(2011\)](#) are achieved. Moreover, the division in the two clouds (active and passive) by using the UV-VJ relation is consistent both with the *SSFR* measured values and with the [Salim et al. \(2007\)](#) division. In the second part, the fits and the evolutions of the active galaxies are coherent with the expected ones from

[Muzzin et al. \(2013\)](#) and [Madau and Dickinson \(2014\)](#).

References

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 "This work is based on observations taken by the CANDELS Multi-Cycle Treasury Program with the NASA/ESA HST, which is operated by the Association of Universities for Research in Astronomy, Inc., under NASA contract NAS5-26555."

Galaxy evolution in dense environments

Evolution of color-mass diagram in density and spectroscopy for a galaxy

Alessia Perego

ABSTRACT

Aims. The aim of the study is to examine the color-mass diagram for a sample in 10,000 square degrees of the sky contained in SDSS survey. In addition, the evolution of NGC 4330 is analyzed.

Methods. Using the spectroscopy and photometry, the galaxies are divided into 4 overdensity bins. They are then plotted in the color-mass diagrams, in which the red and blue sequences are fitted. The fractions of blue, green and red galaxies are obtained and studied as function of the overdensity parameter. For NGC 4330, the evolution of the stripped gas is analyzed by matching the best model to the galaxy spectrum (thanks to the chi square) to recreate the evolution in time of the galaxy.

Results. The expected relations and evolutions are obtained.

Key words. Color, mass, density, environment, evolution

1. Introduction

The role of the environment is fundamental in galaxy evolution: various processes occur when a galaxy lies in an overdens region. These effects are of two main types: gravitational interactions, such as tidal or multiple interactions, and hydrodynamic interactions, as ram pressure, viscous stripping, thermal evaporation and starvation. The combination of internal processes and environmental processes causes an active galaxy to become passive. In a general consideration, this shift from active to passive can be seen in the color mass diagram: there are two main populations, an active one and a passive one, and a transition region, the green valley.

An important contribution to environment studies is given by the Peng et al. (2010)'s observational paper. They identify the features of galaxy evolution (in different masses, environments and redshifts) and interpret them also evaluating their consequences. They found that the overlap of environmental processes and internal processes is in reality completely separable (Figure 1) but both of them concur in making a galaxy passive. These transformations also led to the bi-modality in the galaxy population.

Then the color mass diagram, the different star formation rate SFR and galaxies shape have been studied by Schawinski et al. (2014). They interpret the color mass diagram and the evolution in terms of gas reservoir and how star formation consumes it. In addition, thanks to Galaxy Zoo too, they investigate how the shape of these galaxies change during the evolution. The results found are that all galaxies, even if with different timescales, leave the main sequence, undergo a reduction of SFR and then stabilize in the passive cloud (Figure 2).

Another enhancement in understanding the galaxy evolution was made by Bluck et al. (2019). In particular they analyzed the evolution of the galaxy structure indicated as the ratio between bulge mass and total stellar mass. They confirmed and improved a result of Schawinski et al. (2014): the SFR is connected to galaxy structure.

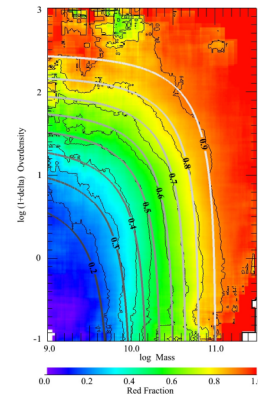


Fig. 1. Figure 6 in Peng et al. (2010): Passive fraction in mass vs overdensity plane.

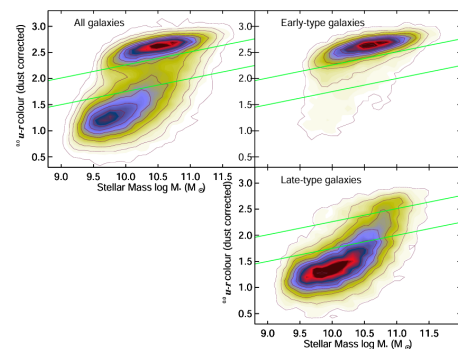


Fig. 2. Figure 3 in Schawinski et al. (2014): color-mass diagram with dust correction.

As regard the study of single galaxies, an important paper is represented by Fossati et al. (2018). They analyzed NGC 4330: thanks to a Monte Carlo code that represents the star formation history (to represent the stripping in the SFHs, the exponential function is truncated), they were able to reconstruct the evolu-

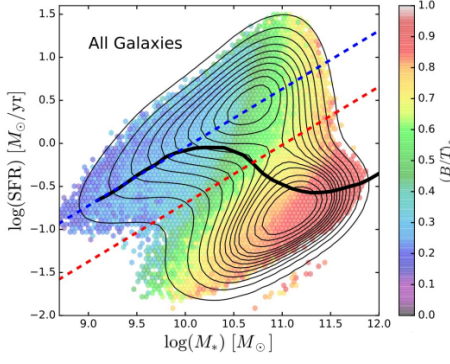


Fig. 3. Figure 1 in [Bluck et al. \(2019\)](#): star-forming main sequence and the passive cloud for SDSS galaxies ($z \sim 0.1$).

tion of the gas stripped from the galaxy. In particular they found a gradient from outside-in (Figure 4): the outer parts had been stripped early on, while the inner ones have not already been stripped totally.

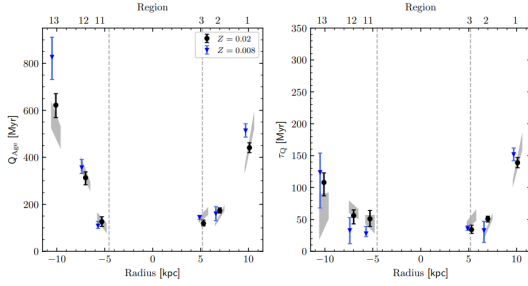


Fig. 4. Figure 8 in [Fossati et al. \(2018\)](#): Quenching age (QAge) vs radius (plus quenching characteristic timescale τ_Q) in the configurations used for NGC 4330.

In this work in particular, environmental effects are studied: firstly how the distribution of galaxies in the color mass diagram changes in different bins of overdensity; and secondly how a specific galaxy is influenced by the environment.

2. Datasets

In the first part the utilized data is from Sloan Digital Sky Survey ([SDSS](#)), based in the Apache Point Observatory, New Mexico. This dataset contains objects from about all the sky in five spectral bands from ultraviolet to the near infrared, both in spectroscopy and photometry. The redshift that the survey is sensible to is $z \sim 0.15$; in this work however, the maximum value considered is $z \sim 0.03$. Another important quantity of the catalog is the local density of galaxies: for each galaxy, a 3D cylinder is considered. They have a base radius of 0.5 Mpc and a redshift depth of $\pm 1000 \text{ km/s}$ and for each of these cylinders, the number of galaxies inside is computed.

The second part is based on the analysis of a Virgo Cluster's galaxy, NGC 4330. This galaxy has an intermediate mass and it's a star forming spiral falling in the center of the Virgo cluster. In this case the data used is from the Very Large Telescope (VLT) of the European Organization for Astronomical Research in the Southern Hemisphere ([ESO](#)) installed in Chile. In particular the instrument used is the Cassegrain FORS2, that, thanks to a slid, it has collected spectroscopic and photometric data from 13 regions of NGC 4330 in a wavelength range of 330-1100 nm.

(Figure 5) In this work, however, the regions considered are only 1, 2, 3, 11, 12 and 13.

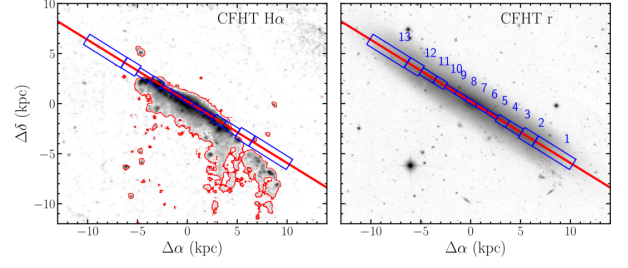


Fig. 5. NGC4330: $H\alpha$ image (left) and r-band image (right).

3. Results

In the first part, the color mass diagram is considered. First of all, to know how many galaxies are present in each region, the mean density of the surrounding for each galaxy is needed. So, from the positions and the redshifts, the volume of observation is computed and from dividing the number of galaxies considered with this volume, the mean density is obtained. Now, the overdensity for each galaxy is calculated by dividing the density value with this total mean density (note: a common rewriting for the overdensity is $\delta_0 = \log_{10}(\delta + 1)$, as it is used in this work). For the color instead, it has been chosen to use g-band and i-band since they are at very different wavelengths (g-band is around 464 nm, and i-band is around 806 nm) and have a good data resolution. From these quantities, the color-mass diagram colored with the overdensity values for all the galaxies is computed (Figure 6). As can be seen, there are two main clouds: one in the upper part, that represents the red sequence and a blue cloud, in the lower part, that contains the star forming galaxies. In the middle there is a less populated region, the green valley.

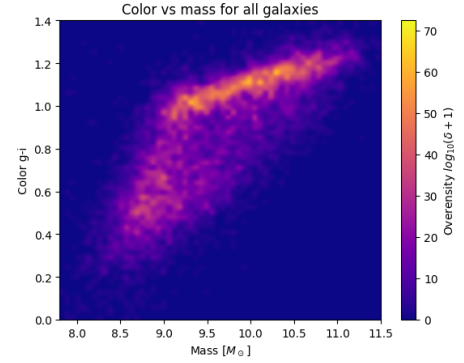


Fig. 6. Color vs mass diagram for all galaxies, colored with overdensity values.

To see how the galaxies are affected by the environment, the color-mass diagram is plotted for 4 different overdensity intervals: $\delta_1=[0, 0.5]$, for isolated galaxies, $\delta_2=[0.5, 2]$, $\delta_3=[2, 2.75]$ and $\delta_4=[2.75, 3.5]$, for the densest regions (Figure 7).

The 4 plots present differences: in the least dens region, the galaxies appear to be mostly in the blue sequence; while as the overdensity parameter increases the red sequence is more and more populated; till the densest situation where all the galaxies are in the red sequence. This shows how the environment influences the star formation of galaxies. A galaxy in a low dens environment is only subject to internal processes, so the star

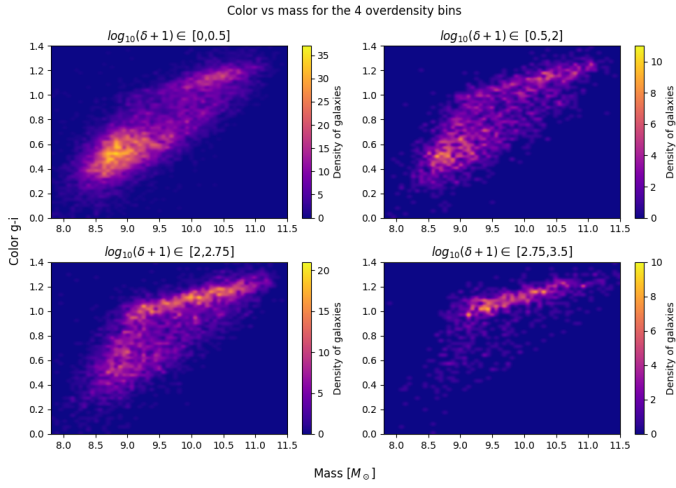


Fig. 7. Color vs mass diagram for the 4 chosen overdensity bins.

formation is less suppressed, and the galaxy can stay active. While in a denser situation, the environmental processes are summed to the internal ones, as obtained by Peng et al. (2010). In this case, the galaxy's star formation is more muffled as the density increase, so from being star forming, the galaxy turns into a passive one and becomes part of the red sequence. The situation in between is called green valley: in here there are galaxies that are coming from the blue sequence and are transforming in passive.

To better see the evolution of how many galaxies there are for each sequence in every overdensity bins, the fraction of galaxies in each situation is calculated. In particular, to obtain a lower limit for the red sequence, the red cloud is linearly fitted and then the line is carried below. While for the upper limit of the blue sequence, the linear fit is done on the blue cloud and then carried up (Figure 8).

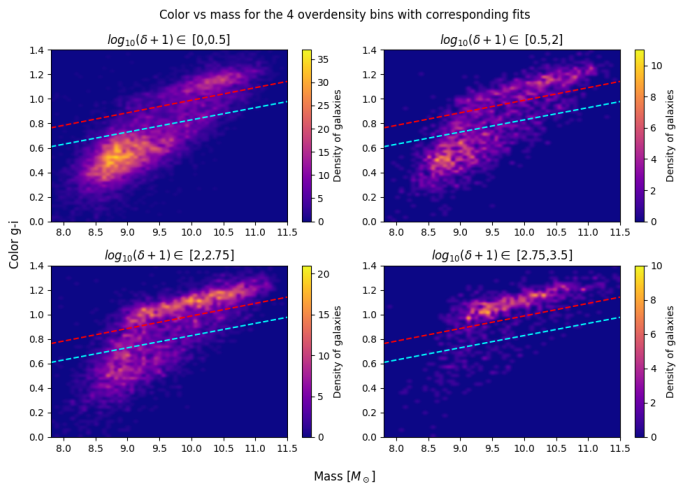


Fig. 8. Color vs mass diagram for the 4 chosen overdensity bins with the fits of red sequence and blue sequence.

From these fits, the three sequences are divided and the fraction of galaxies for each one is computed by dividing the number of galaxies in each cloud with the total number (Table 1). These fractions are then plotted in an histogram to better see the evolution (Figure 9). As expected the number of active galaxies decreases with the increase of the overdensity value, while the red

fraction does the opposite, it grows with bigger densities. The green fraction instead is approximately constant in all the bins.

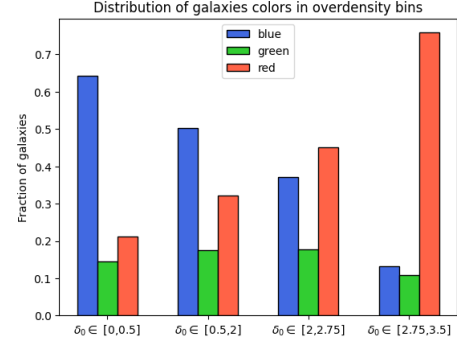


Fig. 9. Histogram of the distribution of galaxies colors for the 4 overdensity bins ($\delta_0 = \log_{10}(\delta + 1)$).

Overdensity	Total	Blu	Green	Red
$\delta_0 \in [0, 0.5]$	10643	6847	1545	2251
$\delta_0 \in [0.5, 2]$	2299	1155	404	740
$\delta_0 \in [2, 2.75]$	5396	2003	959	2434
$\delta_0 \in [2.75, 3.5]$	1006	134	110	762

Table 1. Number of galaxies in each sequence for each bin of overdensity ($\delta_0 = \log_{10}(\delta + 1)$).

In the second part, an analysis of NGC 4330 is done. This galaxy is being stripped by the Virgo's environment, so the spectra reflects this process. The considered regions (1, 2, 3, 11, 12 and 13) have been subjected to stripping at different moments, so by fitting the spectra with a model, the point when the gas has been torn from the galaxy can be found. Each model is a truncated exponential and is characterized by two parameters: the age when the quenching started (Q_{age}) and the characteristic timescale of the decline ($Q\tau$). In spectroscopy however, there are some lines given by the sky distortion, so the spectra considered is only till 5500 nm. This part of the spectra is then fitted with the possible models and the best one is found minimizing the chi square (Figure 10). The best values of Q_{age} and $Q\tau$ are found for each region (Table 2).

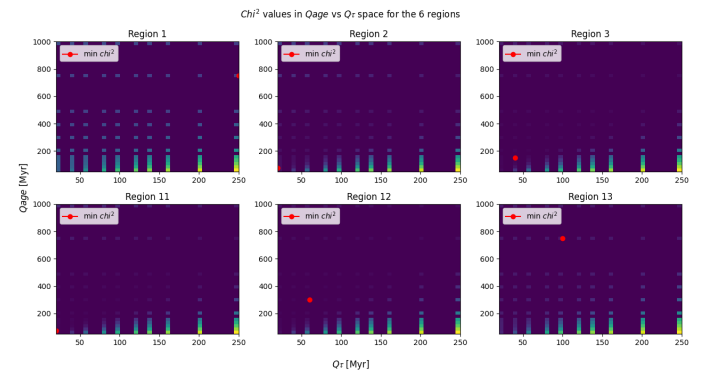


Fig. 10. Histogram of the distribution of the chi square values in Q_{age} vs $Q\tau$ plane.

The values of Q_{age} and $Q\tau$ are then plotted as function of the regions (Figure 11). It is found a gradient outside-in with values similar to the ones in Fossati et al. (2018). Only the value of region 2 doesn't follow the gradient, but on these parameters an error up to 100% can be present. The trend of the data, as expected, can be explained by the stripping and quenching caused by the environment: the outer parts were stripped before then the inner ones, so their value of Q_{age} is bigger. The inner parts are less and later on stripped because they feel a deeper potential given by the galaxy. Another feature that can be seen is that the stripping is not symmetrical because the galaxy is moving inside the Virgo cluster, and so the obtained parameters reflect this behavior (example in region 2 where there's still gas so the best Q_{age} found is smaller to the region 1 value).

Fossati et al., A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) (Astronomy and Astrophysics, 2018)

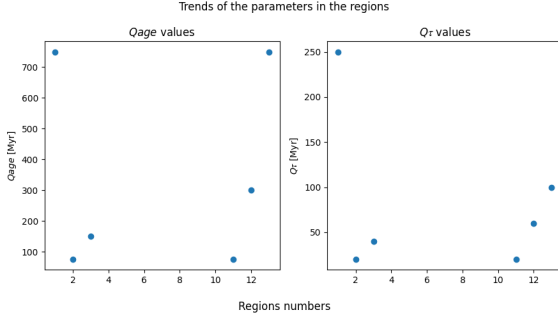


Fig. 11. Q_{age} and $Q\tau$ trends in the considered regions.

Reagion	Q_{age} [Myr]	$Q\tau$ [Myr]
1	750	250
2	75	20
3	150	40
11	75	20
12	300	60
13	750	100

Table 2. Values of the obtained parameters Q_{age} and $Q\tau$.

4. Conclusions

All the results obtained are coherent with the literature. In the first part, the blue and red sequences follow the expected behavior with density, with a bigger blue cloud in underdense regions and a bigger red cloud for the overdense ones. In the second part, the parameters value of Q_{age} and $Q\tau$ are consistent with both with the theory of environment processes and with Fossati et al. (2018) results.

References

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